

MATH3543 Abstract Algebra II

Midterm Exam

October 2026

This exam has three questions. You must complete all three questions to achieve full marks. This exam is worth 30 marks.

Duration: 90 minutes

1. (a) Define *commutator subgroup* and *solvability* of a group. (2)
(b) Find the commutant of the group of symmetric rotations of a regular tetrahedron. (6)
(c) Argue that S_4 is solvable. (2)

2. Let $P = \{1, 2, 3, 4, 5, 6\}$ denote the set of vertices of a regular hexagon. Consider the function
$$\alpha : D_{2 \cdot 6} \times P \rightarrow P : (\sigma, i) \mapsto \sigma(i).$$
(a) Prove that α is a group action. (3)
(b) Find the orbits and stabilisers of this action. (7)

3. (a) State three Sylow theorems. (3)
(b) Describe the isomorphism classes of the Sylow p -subgroups of S_4 . (7)